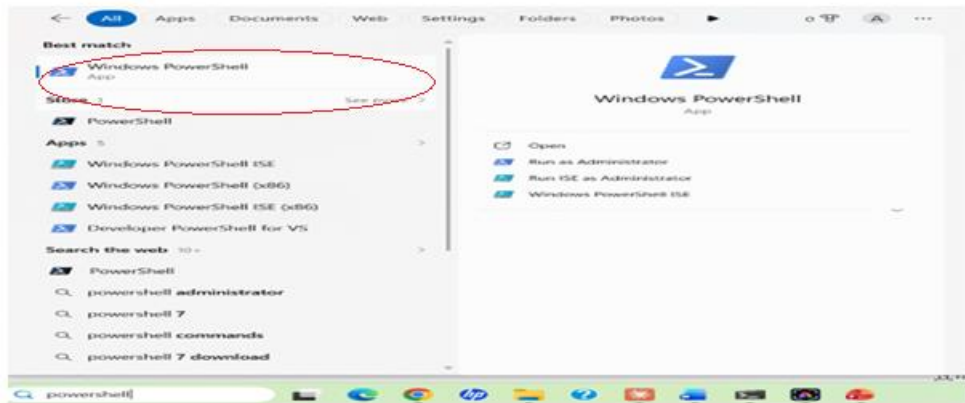
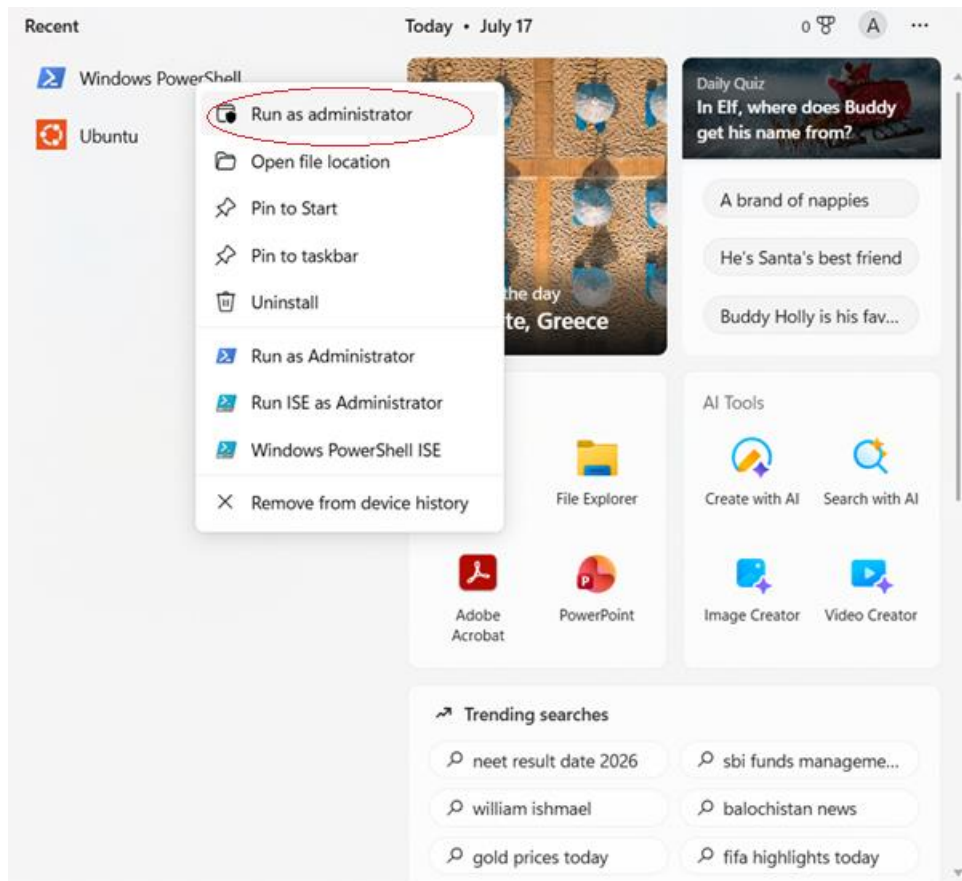


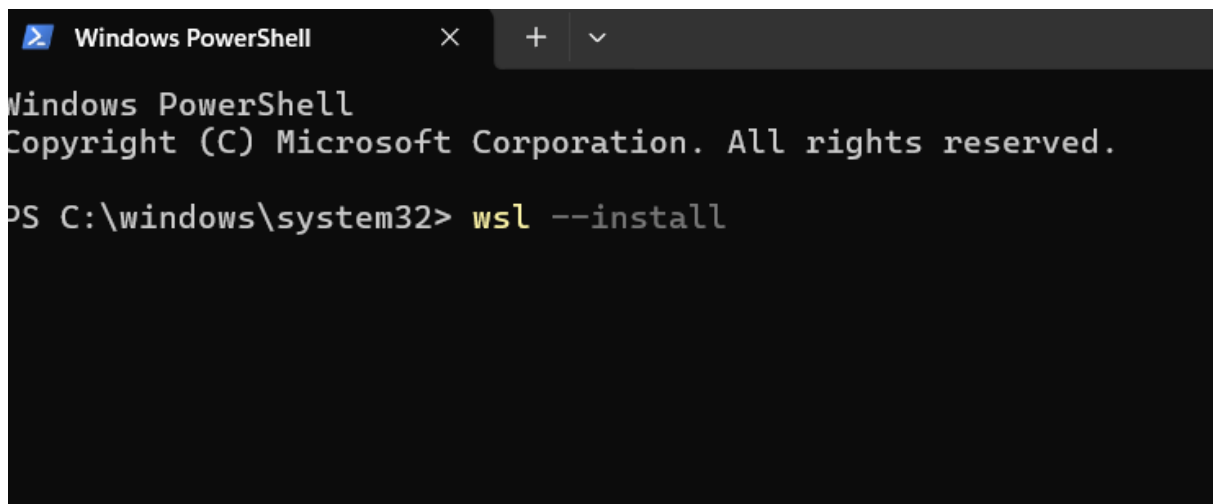
Step 1: Open PowerShell as Administrator

1. Press **Windows**.
2. Search for **PowerShell**.



3. Right-click **Windows PowerShell**.



A screenshot of a Windows PowerShell terminal window. The title bar shows 'Windows PowerShell' with standard window controls. The terminal text reads: 'Windows PowerShell', 'Copyright (C) Microsoft Corporation. All rights reserved.', and 'PS C:\windows\system32> wsl --install'. The 'wsl' command is highlighted in yellow.

```
Windows PowerShell
Copyright (C) Microsoft Corporation. All rights reserved.

PS C:\windows\system32> wsl --install
```

Step 3: Restart Your Computer

After installation, restart your PC when prompted.

Step 4: Complete Ubuntu Setup

After restarting:

1. Ubuntu opens automatically (or search for **Ubuntu** in the Start menu).
2. Wait for the installation to finish.
3. Create a Linux username.

Example:

Enter new UNIX username: anusha

Step 5: Create a Password

You'll then be prompted to create a password.

New password: ****

Retype new password:****

Step 6: Update Ubuntu

```
bhargav08@Bhargava:~$ sudo apt update
[sudo: authenticate] Password:
Get:1 http://security.ubuntu.com/ubuntu resolute-security InRelease [137 kB]
Hit:2 http://archive.ubuntu.com/ubuntu resolute InRelease
Get:3 http://archive.ubuntu.com/ubuntu resolute-updates InRelease [137 kB]
Get:4 http://security.ubuntu.com/ubuntu resolute-security/main amd64 Packages [417 kB]
Hit:5 http://archive.ubuntu.com/ubuntu resolute-backports InRelease
Get:6 http://security.ubuntu.com/ubuntu resolute-security/main Translation-en [97.2 kB]
Get:7 http://archive.ubuntu.com/ubuntu resolute-updates/main amd64 Packages [492 kB]
Get:8 http://security.ubuntu.com/ubuntu resolute-security/main amd64 Components [46.7 kB]
Get:9 http://security.ubuntu.com/ubuntu resolute-security/universe amd64 Packages [159 kB]
Get:10 http://security.ubuntu.com/ubuntu resolute-security/universe Translation-en [51.2 kB]
Get:11 http://archive.ubuntu.com/ubuntu resolute-updates/main Translation-en [118 kB]
Get:12 http://security.ubuntu.com/ubuntu resolute-security/universe amd64 Components [43.2 kB]
Get:13 http://security.ubuntu.com/ubuntu resolute-security/restricted amd64 Packages [349 kB]
Get:14 http://archive.ubuntu.com/ubuntu resolute-updates/main amd64 Components [97.8 kB]
Get:15 http://archive.ubuntu.com/ubuntu resolute-updates/universe amd64 Packages [243 kB]
Get:16 http://security.ubuntu.com/ubuntu resolute-security/restricted Translation-en [67.8 kB]
Get:17 http://archive.ubuntu.com/ubuntu resolute-updates/universe Translation-en [81.0 kB]
Get:18 http://archive.ubuntu.com/ubuntu resolute-updates/universe amd64 Components [184 kB]
Get:19 http://archive.ubuntu.com/ubuntu resolute-updates/restricted amd64 Packages [367 kB]
Get:20 http://archive.ubuntu.com/ubuntu resolute-updates/restricted Translation-en [70.8 kB]
Get:21 http://archive.ubuntu.com/ubuntu resolute-updates/multiverse amd64 Packages [11.3 kB]
Fetched 3170 kB in 5s (676 kB/s)
122 packages can be upgraded. Run 'apt list --upgradable' to see them.
```

Step 7: Install GCC (C Compiler)

```
bhargav08@Bhargava:~$ sudo apt install build-essential
Installing:
  build-essential

Installing dependencies:
  bzip2      g++-15      libalgorithm-diff-xs-perl  libfakeroot      make
  dpkg-dev  g++-15-x86-64-linux-gnu  libalgorithm-merge-perl  libfile-fcntllock-perl
  fakeroot  g++-x86-64-linux-gnu    libcrypt-dev             libstdc++-15-dev
  g++       libalgorithm-diff-perl  libdpkg-perl             lto-disabled-list

Suggested packages:
  bzip2-doc      debian-tag2upload-keyring  g++-15-multilib  bzip      make-doc
  debian-keyring  g++-multilib              gcc-15-doc       libstdc++-15-doc
```

Step 8: Verify the Installation

```
bhargav08@Bhargava:~$ gcc --version
gcc (Ubuntu 15.2.0-16ubuntu1) 15.2.0
Copyright (C) 2025 Free Software Foundation, Inc.
This is free software; see the source for copying conditions. There is NO
warranty; not even for MERCHANTABILITY or FITNESS FOR A PARTICULAR PURPOSE.
```

Step 9: Create Your C Program

Example: nano process.c

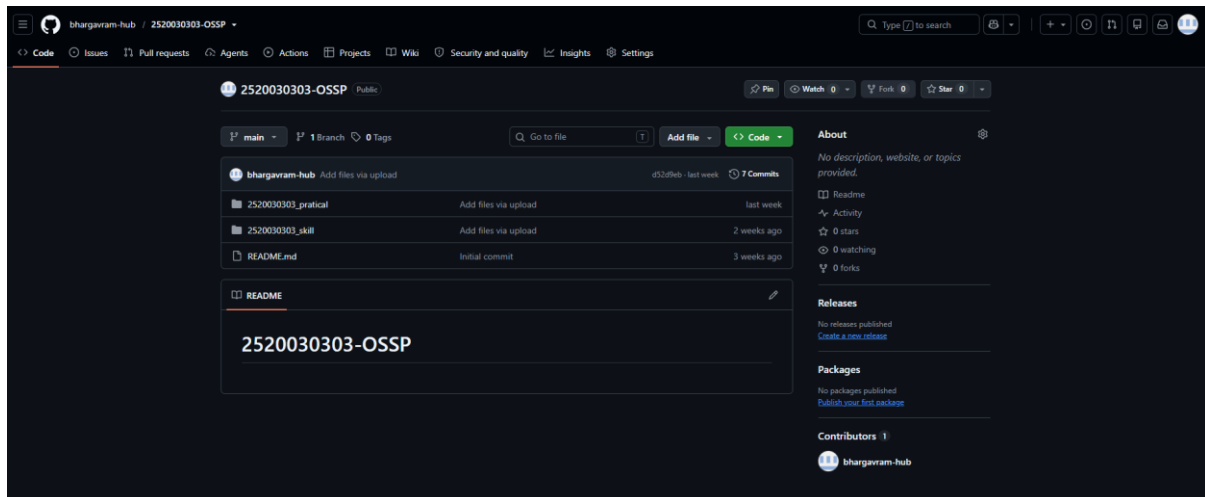
Step 10: Compile the Program

```

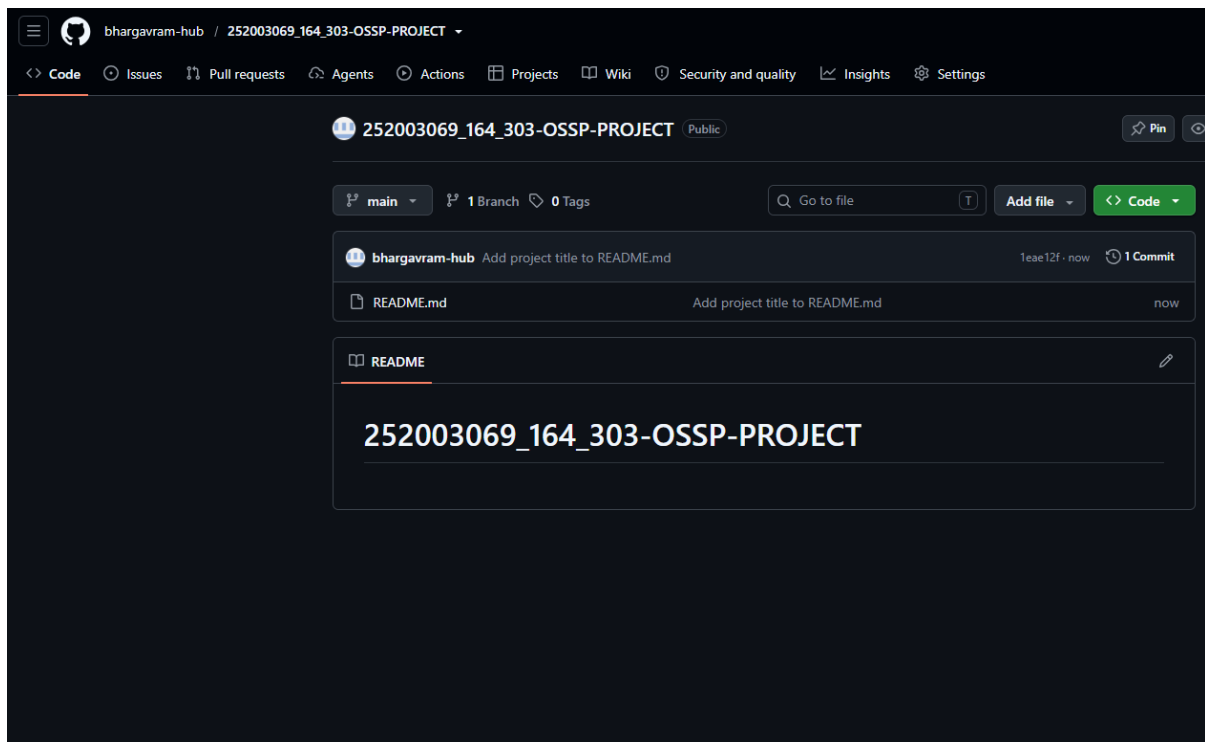
bhargav08@Bhargava:~$ nano hello.c
bhargav08@Bhargava:~$ gcc hello.c -o hello
bhargav08@Bhargava:~$ ./hello
Hello Linux

```

4. GITHUB-Repository :



5. Project Structure :



6. EXECUTION :

1. FORK() :

```
#include <stdio.h>
#include <unistd.h>

int main()
{
    fork();

    printf("Hello\n");

    return 0;
}
```

Output:

```
bhargav08@Bhargava:~$ gcc fork.c -o fork
bhargav08@Bhargava:~$ ./fork
Hello
Hello
```

2. EXEC() FAMILY :

```
#include <stdio.h>
#include <unistd.h>
#include <sys/wait.h>

int main()
{
    pid_t pid;

    pid=fork();

    if(pid==0)
    {
        execlp("date","date",NULL);
    }
    else
    {
        wait(NULL);

        printf("Child Completed\n");
    }
    return 0;
}
```

Output :

```
bhargav08@Bhargava:~$ nano fork_exec.c
bhargav08@Bhargava:~$ gcc fork_exec.c -o fork_exec
bhargav08@Bhargava:~$ ./fork_exec
Mon Aug 24 05:10:51 UTC 2026
Child Completed
```

3. Parent-Child :

```
#include <stdio.h>
#include <unistd.h>

int main()
{
    pid_t pid;

    pid=fork();

    if(pid==0)
    {
        printf("I am Child Process\n");
    }
    else
    {
        printf("I am Parent Process\n");
    }

    return 0;
}
```

Output :

```
bhargav08@Bhargava:~$ nano ParentChild.c
bhargav08@Bhargava:~$ gcc ParentChild.c -o ParentChild
bhargav08@Bhargava:~$ ./ParentChild
I am Parent Process
I am Child Process
```